A project report on Computer Vision Assessment

1. What I Built

Task 1: Basic Image Manipulation

I developed an advanced image processing pipeline using OpenCV, balancing usability, interactivity, and clarity. The pipeline starts with image input handling, then applies a Gaussian blur to simplify intensity-based operations. The interactive Canny edge detection tuner allows users to adjust maximum or minimum threshold values in real time though trackbar. The script saves intermediate and final processed images in an output_task1 directory, and uses matplotlib subplots for better visualization.

Task 2: Mini Project – Color Detector

I developed an object-oriented webcam-based color detection tool using OpenCV. The application features real-time BGR color detection at mouse click positions, with optional color name approximation using Euclidean distance matching against 11 predefined colors. It includes dynamic text color adjustment based on background brightness for better readability, real-time coordinate logging via mouse callbacks, and robust error handling for webcam access.

2. Challenges I faced

To achieve precise edge detection required fine-tuning the Canny maximum and minimum thresholds (optimal range: 34–54), which varied across images, and using a large Gaussian blur kernel (15×15) to reduce noise, though it sometimes blurred important details.

In color matching, using Euclidean distance in BGR space occasionally misclassified similar hues, and webcam auto-white-balance affected color accuracy under varying lighting conditions. Architecturally, adopting an object-oriented approach for Task 2 added complexity but significantly improved code maintainability.

3. Improvements I will do If I have Time

Future updates will include switching color matching to HSV or CIELAB color spaces and implementing dynamic Canny thresholding using Otsu's method. I would have replaced region-based color sampling instead of single-pixel detection for more robust results. Advanced features that I will consider include integrating a machine learning model for more accurate color naming (e.g., trained on Pantone data), adding a webcam calibration routine for consistent color detection, and enabling exportable color palette.